

A Validity Ladder for LLM-Simulated Populations

Patrick Keough

Independent researcher

Author Note

Data, code and every receipt file are at
<https://github.com/pskeough/A-Validity-Ladder-for-LLM-Simulated-Populations>.

Correspondence concerning this article should be addressed to Patrick Keough,
Email: pskeough@gmail.com

Abstract

A simulated sample built by prompting a language model with demographic personas can be used to report a persona’s score, to stand in for individual respondents, to compare groups, to estimate a population’s level or to score a multi-item scale. Each use assumes something different about how well the model replicates people, and validation practice in the field remains uneven. We propose the validity ladder, a gate and four levels released as code, in which every rung tests one of these uses against human data with a statistic and a pass rule, and a sample supports only the claims whose rungs it passes. In a worked example of 28,800 PHQ-8 depression assessments from four models, compared against a national health survey, a few repeated draws make each persona’s mean readable. Above the gate every model fails, with answers less varied than people’s, an exaggerated income gradient, every group placed too high and a distorted scale structure. Real survey respondents put through the same design are rarely failed, while failures planted into them are caught at the level each targets. Tested on three published silicon samples, the ladder supports only a few partisan gaps, pointing toward narrower readings of those samples’ claims. These results show that each claim drawn from a simulated sample needs its own check against the human data it claims to represent.

Keywords: silicon samples, large language models, simulated participants, validity, measurement invariance, PHQ-8, NHANES

A Validity Ladder for LLM-Simulated Populations

Behavioral researchers now prompt large language models (LLMs) with demographic personas and treat the answers as *silicon samples*. Argyle et al. (2023) used such samples to reproduce vote choice and opinion by group, Park, Schoenegger, and Zhu (2024) used them to replicate classic experiments, and Lin (2026a) describe their use in piloting instruments before fielding them. For a study, a simulated sample costs little, reaches groups that are hard to recruit, and can be rerun whenever the design changes. Each of these uses assumes that the simulated sample stands in for a real one in the respect the study measures.

In practice the stand-in assumption is checked unevenly. Across eight social science journals, Desai, Card, and Jacobs (2026) found validation practice in LLM studies “inconsistent and limited.” Park et al. (2024) found GPT-3.5 answering with almost no variation where people vary, and Bisbee, Clinton, Dorff, Kenkel, and Larson (2024) found synthetic opinion distributions narrower than the survey distributions they imitate. Lin (2026b) proposed a workflow in which the validity a study needs rises with the claim it makes. That workflow does not yet supply the tests themselves, each with a pass rule and a population reference, in an order a researcher can run on one model and report.

One persona from the corpus analysed here shows what reading individual answers misses. Prompted as a single Black woman on Medicaid with a household income under \$35,000, GPT-4o-mini returns a PHQ-8 total of 8 on its first draw, a plausible answer in the mild band. Over thirty draws the same prompt averages 10.3. Adults in the National Health and Nutrition Examination Survey (NHANES) who share her race, sex, income and marital status average 4.7. None of her single answers would show that the simulated population she belongs to sits well above the real one.

We propose a protocol for the missing check, the validity ladder (Figure 1). A gate first measures how many repeated draws make a persona’s score precise enough to read. Above the gate, level 1 tests whether a single draw is an answer pattern a person could

give, and level 2 tests whether gaps between groups match the population's. Level 3 tests whether the population level matches, and level 4 tests whether the items relate to one another as they do in people. Each rung of the ladder, the gate or one of the levels, carries a statistic, a pass rule and the use a pass supports, and a simulated sample supports only the claims whose rungs it has passed. With the rules frozen and released as code, a researcher can run the ladder on their own sample and report the verdicts.

The worked example applies the ladder to 28,800 PHQ-8 depression assessments from GPT-4o-mini, Gemini-3-Flash, DeepSeek-V3 and GLM-4.7, a corpus first collected for an earlier audit (Keough, 2026) and rebuilt here from the raw answers. A few repeated draws suffice to read each persona's mean, yet no model passes any level above the gate. Single answers vary less than people's, and the models exaggerate the income gradient, place every group's level too high and distort the scale's structure. Before these failures can be credited to the models, the ladder has to be shown not to fail real people. Survey respondents drawn through the same design rarely fail any level, and failures planted at known sizes are caught by the level built to find them.

Three published silicon-sample datasets test whether the ladder travels beyond one instrument. Feeling thermometers from Bisbee et al. (2024), vote choice and attitudes from Argyle et al. (2023) and opinion distributions from Meister, Guestrin, and Hashimoto (2025) go through the same frozen rules. The ladder keeps a few partisan gaps and leaves most other gaps failed, unresolved or unreadable, pointing toward narrower readings of each release's claims.

The Validity Ladder

A researcher who prompts a language model with personas and collects its answers to a fixed instrument has a simulated sample, and the validity ladder is a set of checks on what that sample can be used to claim. The instrument may be a multi-item scale, a single survey item or an opinion question. Each call to the model returns one draw, and each persona under one model and one prompt framing forms a cell of k draws whose average is

the persona mean. Reading a persona’s score, treating one answer as a case, comparing groups, estimating a group’s level and scoring a scale are separate claims, and each rests on a different property of the sample. Ordering these claims from narrow to broad, the ladder begins with a gate that tests the base premise, a score stable enough to read, and each level above the gate supports a broader claim that the sample must earn on its own. Without these checks, a researcher cannot tell answers that only look coherent from coherent ones, or a sample that looks typical from one free of bias.

Using the ladder starts from critical doubt about whether the model can stand in for the people it simulates, and from the human data that can test it. Before generating anything, the researcher identifies which reference can ground the sample and which claims about the population the study intends to make. The rungs differ in what they need (Figure 1). Repeated draws of each persona are enough for the gate. Levels 1 and 4 require a multi-item scale and individual human answers on it, while levels 2 and 3 require personas built on attributes the reference records and a reference that measures the resulting groups precisely. A single-item outcome such as a feeling thermometer or a vote therefore supports the gate and levels 2 and 3. Where a study cannot meet a rung’s requirements, that rung is not run and its claims are not made.

Verdicts are given per model, because a study deploys one model and a pooled verdict can hide errors running in opposite directions. Passing a test means it found no failure at the precision the data allow. When the reference is too imprecise to support any verdict, the test is recorded as not read. The gate sets k for levels 2 and 3, and a failure at level 1 rules out only single-draw uses, leaving the remaining levels to run. Only the gate is a prerequisite, and the levels above it are ordered by how broad a claim each one supports. Group comparisons of a multi-item total additionally assume that the scale measures the same construct in each group, making a pass at level 2 or 3 on a scale total valid only alongside a pass on R4 at level 4. All rules and thresholds were fixed before the three external datasets were analysed, and they serve as defaults a user may change. Supplement

S7 gives the full specification, with the estimators, bootstrap designs and multiplicity control.

Figure 1

The Validity Ladder

	Rung and question	A pass lets you claim	A failure looks like	Your study needs
↑ broader claims	LEVEL 4 Structural fidelity Do the items form the same scale as in people, in every group?	The total can be scored as one score and compared across groups.	The total stops measuring one thing, or measures it differently by group.	A multi-item scale, individual human answers and enough personas per group.
	LEVEL 3 Population calibration Is each group's level right?	A group's simulated level (mean, prevalence, share agreeing) can stand in for the real one.	The whole sample is shifted, or a right mean hides a wrong prevalence.	Human levels for the same groups, and weights to match the persona mix.
	LEVEL 2 Subgroup fidelity Are the gaps between groups the right size and direction?	Group differences can be read from the sample: an income gradient, a partisan gap.	A gap is exaggerated, shrunk, erased or reversed.	A human reference that measures the gap precisely, for groups the personas match.
	LEVEL 1 Individual coherence Could a real person give this answer, and do answers vary as much as people's?	Single draws can serve as individual cases: vignettes, test items, synthetic respondents.	Every persona answers alike, or answer patterns are too regular to be human.	A multi-item scale and individual human answers on it.
	GATE Regeneration stability Is a persona's score stable enough to read?	The average of k draws can be read as the model's answer for that persona.	Answers change from draw to draw, so a single draw is mostly noise.	Repeated draws of each persona. Any outcome.

Note. Read from the gate at the bottom to level 4 at the top. Each rung is read for each model on its own. A level-1 failure rules out single-draw uses only, and levels 2 to 4 still run. A rung whose requirements a study cannot meet is not run, and the claims it protects are not made. The statistic and pass rule for each rung are given in the text, and Supplement S7 gives the full specification.

Gate: Regeneration Stability

Any study that reports a persona's score, or builds a claim on one, depends on the gate. Two calls to a model with the same persona prompt rarely return the same answer,

because the model samples each response. A study that reads one generation as a persona's answer therefore reads the persona's score plus noise, and the gate asks how many draws must be averaged before that noise is small enough for the score to be read. Following a generalizability study (Brennan, 2001; Cronbach, Gleser, Nanda, & Rajaratnam, 1972), the variation splits into σ_p^2 , the spread between personas, and σ_r^2 , the spread among draws of one persona. Averaging k draws gives a standard error and a dependability of

$$\text{SE}(k) = \sqrt{\sigma_r^2/k}, \quad \phi(k) = \frac{\sigma_p^2}{\sigma_p^2 + \sigma_r^2/k}. \quad (1)$$

The standard error is the typical distance between a persona mean and the model's long-run score. Dependability, between 0 and 1, is the share of variation among persona means owed to real differences between personas.

Setting the pass rule in units of the reference standard deviation lets the same rule carry across instruments. Means of k draws may be read once $\text{SE}(k)$ falls to 0.25 reference standard deviations in each prompt framing, and the recommended k takes it to 0.125. Single draws may be read only if the draw-to-draw standard deviation is itself no larger than 0.25 reference standard deviations. Dependability appears beside every verdict but is required ($\phi \geq .80$) only when a study compares or ranks individual personas, because a faithful simulator reproduces the small differences between demographic cells that real populations show. Passing the gate supports reading the k -draw mean as the model's answer for that persona, leaving the correctness of that answer to the levels above. Failing it rules out persona-level readings until more draws are averaged.

Level 1: Individual Coherence

Uses such as exemplar vignettes, test items for a screener and synthetic respondents treat single draws as individual cases. Real people with the same total answer in different patterns, most endorsing the common items before the rare ones and a minority answering in unusual ways, while a simulator can return each total in the same textbook pattern.

Level 1 tests whether single draws look like answers real people give and whether they vary

as much as people do.

For the test, a graded response model (Samejima, 1969) is fitted to the human reference with survey weights, and every respondent and every draw receives the standardised person-fit statistic l_z^* (Drasgow, Levine, & Williams, 1985; Sinharay, 2016; Snijders, 2001), low for an unusual pattern and high for an unusually regular one. Each draw is then set against real respondents with the same total, and two ratios summarise the comparison. The *misfit ratio* divides the share of draws more unusual than 95% of matched respondents by the 5% expected, and the *overfit ratio* does the same for draws more regular than 95% of them. In a real population both equal 1, and a value of 0.2 means the model produces a fifth as many such patterns as people. A model passes in a framing when the 90% intervals of both ratios fall between $1/\tau$ and τ . The tolerance τ is the smallest of 1.5, 2 and 3 that covers every clear departure from 1 shown by real subgroups of the reference scored against the whole, and it never drops below the 1.5 floor set by the positive control in the Controls section.

A model with too few patterns in both tails is *compressed*. Passing supports single draws as individual cases, and a compressed failure rules out any use resting on the spread of real presentations.

Level 2: Subgroup Fidelity

Many persona studies are built to compare groups, reporting an income gradient in a symptom score, a partisan gap in feeling toward a party or a gender gap in an attitude. Level 2 asks whether the gap between two groups has the population's size and direction. The simulated gap g compares personas that differ on one attribute and match on the rest, such as two personas identical except for income. In the human reference, the population gap γ is computed the same way among people who match on the other attributes, and the crude reference gap is reported beside it. Level 2 reads the ratio $\rho = g/\gamma$, where 1 reproduces the population gap, 2 doubles it, 0 erases it and a negative ratio reverses it. We compute the interval for ρ as a Fieller interval, the standard interval for a ratio of two

estimates (Fieller, 1954), widened for the number of contrasts tested on each model.

A contrast is KEPT when its whole interval lies between 0.75 and 1.25, placing the simulated gap within a quarter of the real gap’s size. When the interval lies wholly outside that range the contrast is NOT KEPT, and when it overlaps the range the contrast is UNRESOLVED. For a contrast that is not kept, the ratio names the direction of the error, with ratios above 1.25 steepened, ratios from 0.25 to 0.75 attenuated, ratios near zero missing and negative ratios reversed. Before any verdict, a contrast is set aside as NOT READ in two cases. In one the reference cannot tell its population gap from zero, and in the other the reference measures the gap too imprecisely for any simulated gap to be certified kept, unless the contrast is already clearly not kept.

A model passes level 2 when every contrast read is kept, fails when any is not kept, and is otherwise unresolved. A pass supports reading the size and direction of group differences from the simulated sample, leaving either group’s level to level 3. A failure rules out the group comparisons it names.

Level 3: Population Calibration

Level 3 serves studies that use a simulated sample in place of a survey to estimate a group’s level, such as the prevalence of a condition, a group’s mean attitude or the share agreeing with a policy. Level 2 compares groups with each other, and level 3 compares each simulated group with the real group it stands for. Before that comparison the persona grid has to be weighted like the population. Designed rather than sampled, a grid holds one persona for each combination of attributes, while a real group contains those combinations in unequal shares. Each persona cell c in a group G is therefore weighted by its share p_c of that group in the reference, and the residual for the group is

$$R_G = \sum_{c \in G} p_c (s_c - y_c), \quad (2)$$

where s_c is the persona mean and y_c the mean of real respondents who match that persona (Holt & Smith, 1979). A residual of +2 places the simulated group 2 points above the real

group on the instrument. Draw error within cells and a design-based jackknife of the survey together give its interval.

Passing is an equivalence test. A group row passes at tolerance δ when its 90% interval lies within $\pm\delta$, tested by two one-sided tests (Lakens, Scheel, & Isager, 2018; Schuirmann, 1987), with δ named by the researcher as the smallest difference that matters for the use. Half a reference standard deviation is the default and 0.2 standard deviations the strict alternative. Every group row must pass for the model to pass. Because a correct mean can hide a wrong prevalence, the share at or above the instrument's cut point enters as a second row. A pass supports using the simulated level in place of the group's level, within δ , for the prompt, persona grid and survey period tested. A failure rules out reading the sample as an estimate of that level.

Level 4: Structural Fidelity

Level 4 serves studies that add the items of a multi-item scale into one total, such as a symptom score or a personality trait score, and compare groups on that total. Because the items of a validated scale rise and fall together in people, the total measures one thing, and measures it the same way in every group. A simulator can return totals that look right while its items move together differently from people's, and the total then loses the meaning it has for people. Level 4 treats the draws of one model in one framing as one population, resampled by persona, and fits a one-factor model to the polychoric item correlations (Olsson, 1979). The fit is checked against three rules, numbered as in the full specification in Supplement S7. Rule R1, a general factor, requires a single dimension behind the items, with every loading at least .30 and a first eigenvalue at least three times the second, or the reference's own value where the reference falls short. Rule R2, loadings like people's, requires the items to load on that factor with the pattern and strength seen in people, read as Tucker's congruence of at least .95 (Lorenzo-Seva & ten Berge, 2006) and a root mean square loading difference of at most .10, each at the conservative end of its 90% interval. Rule R4, the same scale in every group, requires the factor to work the same

way across the groups the reference records wherever it does in people, read on the RMSEA of each invariance step (Savalei, Brace, & Fouladi, 2024) against a margin of .08.

A model passes level 4 when it passes all three rules in every framing. R1 and R2 together support scoring the total as one score with item weights like people’s. R4 adds that a group comparison of the total carries the same meaning as in people, and passes at levels 2 and 3 are read beside it. The specification’s R3 is a diagnostic and does not decide the verdict. It reports whether the factor holds among draws of one persona, as it does among people who share a demographic profile.

Worked Example

Data

The worked example uses the corpus of Keough (2026), in which language models answered the eight-item Patient Health Questionnaire depression scale (PHQ-8; Kroenke et al., 2009) as demographic personas. Four models, GPT-4o-mini, Gemini-3-Flash, DeepSeek-V3 and GLM-4.7, generated the corpus through OpenRouter between 28 December 2025 and 11 January 2026. The persona grid crosses race (Asian, Black, Hispanic, multiracial, White), gender identity (cisgender and transgender women and men), income (three levels) and relationship status (married, single), for 120 personas. Income was stated as a dollar-and-insurance label, from “Low (<\$35k, Medicaid)” to “High (>\$250k, Concierge)”. Each persona answered 30 times under each of two framings, one listing its attributes as fields (clinical) and one giving them as a first-person self-description (narrative). A shared system prompt cast the model as a “Clinical Simulation Engine” answering from “the statistical likelihood of symptoms for this specific demographic intersection in the US population”. With no sampling parameter set, every draw ran at the provider’s default decoding. The corpus holds 28,800 assessments, of which 28,799 have a valid PHQ-8. Supplement S1 prints the prompts, Supplement S3 gives the provenance of every row, and the generation code is released with the data.

For the population reference we use the 36,274 adults in NHANES 2005–2018 who

answered all eight PHQ-8 items, analysed with the survey’s weights and design (Chen et al., 2018; Johnson et al., 2013; National Center for Health Statistics, 2018). Income is the ratio of family income to the poverty line, banded below 1.3, from 1.3 to 3.5, and above 3.5, and each persona’s income label is mapped to one band. Forty-eight of the 120 personas have a matching group in the survey. Because NHANES records sex rather than gender identity and does not identify multiracial adults on their own, transgender and multiracial personas enter levels 1 and 4, which read the whole simulated population, and are left out of every group comparison. NHANES 2021–2023 and other reference choices are sensitivity analyses in Supplement S3.

On this corpus the NHANES standard deviation of the PHQ-8 total sets the gate tolerances at 0.98 and 0.49 points and the default level-3 tolerance at about 2 points. Scored against the whole survey, no NHANES subgroup by sex, race and ethnicity, income or survey cycle departs from 1 by more than a factor of 1.36, so level 1 uses $\tau = 1.5$. Level 2 reads seven contrasts for each model, the sex gap, three income gaps and three racial gaps, with Bonferroni-adjusted 98.6% intervals. Levels 2 and 3 use the mean of all 30 draws of each persona, more than the gate requires for any model, and level 4 tests invariance across sex, race and income.

Results

Table 1*Ladder Verdicts per Model in the Worked Example*

Model	Gate: draws needed (min. / rec.)	Level 1: single draws	Level 2: income gap ratio	Level 3: overall residual	Level 4: structure
DeepSeek-V3	Pass (3 / 9)	Fail, compressed	Fail, 1.91	Fail, +4.6	Fail, no general factor
Gemini-3-Flash	Pass (3 / 10)	Fail, compressed	Fail, 5.07	Fail, +2.6	Fail, loadings (R2) and invariance (R4)
GPT-4o-mini	Pass (4 / 13)	Fail, compressed	Fail, 2.16	Fail, +4.7	Fail, no general factor
GLM-4.7	Pass (6 / 21)	Fail, too few atypical	Fail, 5.03	Fail, +2.1	Fail, loadings (R2) and invariance (R4)

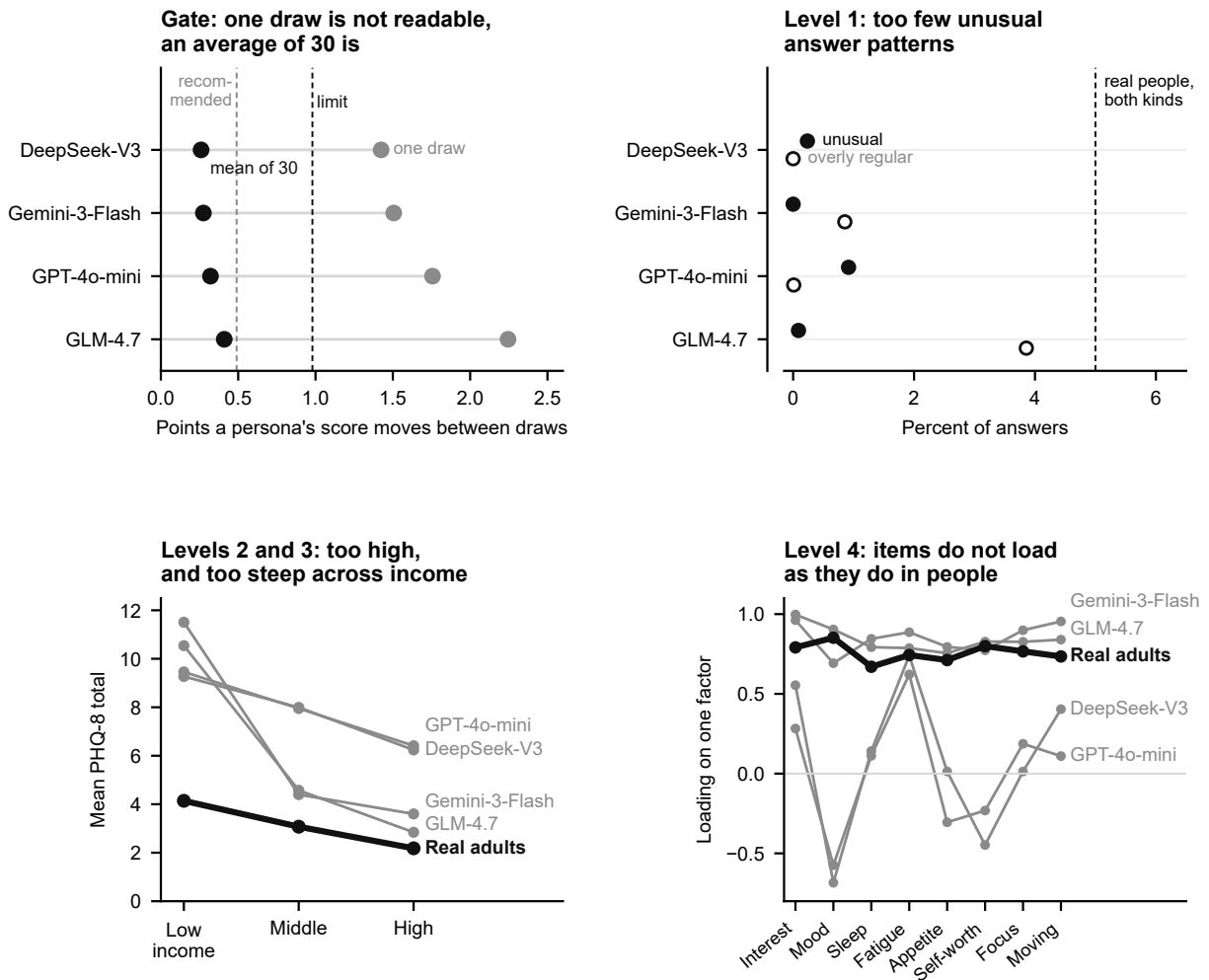
Note. Gate: draws needed for a persona mean's standard error to reach 0.25 (minimum) and 0.125 (recommended) NHANES standard deviations, the larger of the two framings. Level 2: ratio of the simulated to the standardised NHANES gap between low- and high-income personas, where 1 reproduces the gap. Level 3: post-stratified simulated minus NHANES mean PHQ-8, in points. Level 4: R2, loadings stronger than people's; R4, the factor differs across income where it does not in NHANES. Full per-level tables are in Supplement S4.

Table 1 gives every model's verdict at each rung, and Figure 2 shows what each failure looks like. All four models pass the gate and fail every level above it.

Gate. No model can be read from a single draw. The draw-to-draw standard deviation is 1.25 to 2.24 points in every model and framing, above the 0.98-point tolerance.

Figure 2

What Each Failure Looks Like in the Worked Example



Note. Gate: how far a persona's PHQ-8 total moves between draws, for one draw (the draw standard deviation, larger framing) and for the mean of 30 (its standard error), against the gate's limit of 0.98 points and recommended level of 0.49. Level 1: percent of draws more unusual (filled) or more regular (open) than 95% of NHANES adults with the same total, both framings, where real people give 5% of each. Levels 2 and 3: post-stratified mean PHQ-8 total in each income band. Level 4: loading of each PHQ-8 item on a single factor in the narrative framing. DeepSeek-V3 and GPT-4o-mini have no general factor, and Gemini-3-Flash and GLM-4.7 follow the human pattern with loadings stronger than people's. Every level-2 interval is in Supplement Figure S1.

Averages of 30 draws pass in every model, and the minimum rule needs 3 to 6 draws. Two draws of one persona fall in different PHQ-8 severity bands 35.2% of the time [33.6, 36.8]. In a separate decoding control on 12 personas, setting GPT-4o-mini’s temperature to 0 cut its draw variance to 12.5% of the default without separating personas any better (Supplement S1.10). Every reading that follows therefore uses averaged draws.

Level 1. No model passes level 1, in either framing or at any tolerance up to 3. At matched totals NHANES puts 5% of adults in each tail of person fit, and the models put 0.0% to 1.0% in the unusual tail and 0.0% to 4.3% in the regular tail. Three models are compressed, and GLM-4.7 is short of unusual patterns alone. The models also weight depressed mood more heavily than people do, scoring 0.16 to 0.51 points higher at matched totals on depressed mood and 0.22 to 1.11 points lower on fatigue, one of the physical symptoms people report first. They reuse answer sheets as well, giving two different personas the identical answer vector 4.6 to 39 times as often as two NHANES adults with the same total. Any use that treats a draw as a person, from a vignette bank to a test set for a screener, would draw from a pool missing almost all of the unusual presentations that one in twenty real respondents give.

Level 2. Every model fails level 2 on income, and Supplement Figure S1 gives every interval. The gap between low- and high-income personas is 1.91 (DeepSeek-V3) to 5.07 (Gemini-3-Flash) times the population gap. Of the 12 per-model income contrasts, 10 are steepened, and the survey measures the other two gaps too imprecisely to certify kept. We read the size of the low-to-high ratio as partly a product of how the persona’s income label is matched to the survey. Matching on the dollar amount in the label (Supplement S3) brings the ratio down to 1.04 for DeepSeek-V3 and 1.18 for GPT-4o-mini, which leaves both contrasts unresolved, and Gemini-3-Flash and GLM-4.7 stay steepened at 2.76 and 2.74. Under that matching DeepSeek-V3 and GPT-4o-mini still steepen the middle-to-high gap (3.29 and 2.89), leaving every model failing level 2. On the sex gap the models disagree, with GPT-4o-mini and DeepSeek-V3 reproducing a quarter to two fifths of it

(ratios 0.23 and 0.41), GLM-4.7 doubling it (2.06) and Gemini-3-Flash unresolved. Pooled over the four models the ratio is 0.97, a near match built from errors in opposite directions. No racial contrast is kept in any model. Among adults matched on income, sex and marital status, the survey cannot tell the Hispanic–White gap from zero and measures the Black–White gap too imprecisely to certify kept, and GPT-4o-mini’s Black–White gap is still not kept. For a researcher, these verdicts rule out estimating from any of these samples how depression differs by income, sex or race. Because the survey measures the Black–White and Hispanic–White gaps too imprecisely to certify any simulator, a study of those gaps needs a larger or more targeted reference before the ladder can read it.

Level 3. Every simulated population is more depressed than NHANES. After reweighting, the overall residual is +2.1 points for GLM-4.7, +2.6 for Gemini-3-Flash, +4.6 for DeepSeek-V3 and +4.7 for GPT-4o-mini. All 40 model-by-group residuals are positive, and no model passes all ten groups at 2 points. In every model the largest residual is in the low-income band. The failure holds under all eight reference specifications, including NHANES 2021–2023, and only whether the national row passes depends on the reference chosen (Supplement S3). Prevalence, the share scoring 10 or more, errs in a different pattern from the mean. In the low-income band 55.5% of simulated respondents score 10 or more, against 13.6% of real adults. Across the middle and high bands the simulated share is too low although the mean is too high, and DeepSeek-V3’s overall prevalence looks correct only because its low-income excess cancels these shortfalls. Read as a stand-in for a survey, these samples would overstate depression for every group, most of all among low-income adults.

Level 4. No model passes level 4. DeepSeek-V3 and GPT-4o-mini have no general factor, with first-to-second eigenvalue ratios of 1.09 to 1.73 against 7.47 in NHANES. Gemini-3-Flash and GLM-4.7 have a factor with the human pattern of loadings (congruence .990 to .998), and the loadings are stronger than people’s. Gemini-3-Flash fails R2 in both framings and GLM-4.7 in the narrative framing, and both fail invariance across

income, which holds in NHANES. In these two models 48% to 64% of item variance lies between personas, against 3.5% between demographic cells in NHANES, and draws of one persona share no general factor. Built from differences between persona means, their factor rests on the same uniformity within personas that level 1 found. For a researcher, the PHQ-8 total from DeepSeek-V3 or GPT-4o-mini cannot be scored as a single measure, and in Gemini-3-Flash and GLM-4.7 a comparison of the total across income levels lacks the meaning it carries in NHANES.

One persona. Followed through the ladder, the introduction’s persona carries each verdict on a single case. She is a single Black woman on Medicaid with an income under \$35,000, and GPT-4o-mini gives her 30 clinical draws ranging from 7 to 16. Her first draw of 8 lands in the mild band and her mean of 10.3 in the moderate band, so her readable score is the average. Her 60 draws are each plausible as human answers and together less varied than real people at her severity, repeating the level 1 failure. Her mean over both framings, 10.0, exceeds the 4.7 of matched NHANES adults by 5.3 points, repeating the level 3 excess. Level 2 reaches her only through the gaps her groups form with others.

Summary. Read one at a time, every model’s answers look like believable patients, and averages of a few draws are stable enough to read. A researcher who checked this sample by reading its answers, or by the stability of its scores, would have accepted it. Above that stable score the ladder fails the same sample on every use, finding its individual answers too uniform, its income gradient exaggerated, every group too depressed and its scale total without the structure the PHQ-8 has in people.

What this example does not show. These verdicts describe four base models under one prompt family, with one wording per attribute, one persona per demographic cell, no age in the personas and the PHQ-8 items in a fixed order. The shared system prompt asked for the statistical likelihood of symptoms for each persona. Removing that instruction did not remove the excess at level 3. When a prompt control on 12 personas (Supplement S1.9) asked each model to answer as the person described, DeepSeek-V3 and

Table 2*Claims a Study Might Make From This Corpus, and What the Ladder Allows*

Rung	Claim	On this corpus
Gate	A persona’s score can be read from one response.	No. Two draws of one persona fall in different severity bands 35.2% of the time.
Gate	The average of 30 responses is the model’s score for that persona.	Yes, for all four models, which need 3 to 6 draws under the minimum rule.
Level 1	A single response can stand in for a real patient.	No. Unusual answer patterns are 0.0% to 1.0% of draws, against 5% of people.
Level 2	Low-income adults score higher than high-income adults, by this much.	Direction only. The gap is 1.91 to 5.07 times its size in the population.
Level 2	Women score higher than men, by this much.	No. Two models shrink the gap, one doubles it and one is unresolved.
Level 2	Black and White adults differ by this much.	Not certifiable. The survey measures this gap too imprecisely to certify any model.
Level 3	This share of low-income adults is depressed.	No. 55.5% of low-income simulated respondents score 10 or more, against 13.6% of real adults.
Level 3	Depression in the population is this high overall.	No. Every model is 2.1 to 4.7 points too high overall.
Level 4	The PHQ-8 total is one score that means the same in every group.	No. Two models have no general factor, and in the other two the factor differs across income.

GPT-4o-mini scored 5.8 to 7.4 points above NHANES in every income band, compared with 4.0 to 5.3 under the corpus prompt, while Gemini-3-Flash’s low-income excess grew

from 8.9 to 10.0 points. Each income label also paired a dollar amount with an insurance type, a cue the prompt control left unchanged. Transgender and multiracial personas have no NHANES comparison group and were not tested at levels 2 and 3. A different prompt, a persona grounded in real respondents or a fine-tuned simulator could pass where these models fail, and running the ladder on that sample would show whether it does.

Controls: Real People and Planted Failures

With every model failing every rung above the gate, the ladder itself has to be shown to leave real people unfailed and to catch failures of a known size. The controls build pseudo-models out of real survey respondents. Each replicate splits the NHANES adults at random into two halves within the 48 persona cells, uses one half as the reference, and fills the corpus design from the other half, with 120 personas, two framings and 30 draws, where each draw is a real respondent from the persona's cell. A panel of four such pseudo-models stands in for the four language models. Failures are then planted into the same respondents, each aimed at one rung, by making the items independent, shifting every persona mean, doubling the income gap, compressing answer patterns toward the typical one, and making three items independent within the low-income group. Run on the real corpus with the full reference, the control code reproduces all 303 published rung numbers.

Table 3*Verdicts for Seven Simulator Types Built From Survey Respondents*

Simulator	Gate	L1	L2 fail	L3	R1	R2	R4 fail	R4 unres.	L4
Real respondents	100.0	92.9	0.0	100.0	100.0	100.0	4.0 (25)	96.0	0.0
Independent items	100.0	0.0	0.0	100.0	0.0	0.0	0.0 (5)	0.0	0.0
Shifted, +2.1	28.6	72.9	0.0	0.0	100.0	100.0	0.0 (5)	100.0	0.0
Shifted, +4.7	0.0	15.7	0.7	0.0	100.0	100.0	0.0 (5)	100.0	0.0
Income gap doubled	100.0	84.3	92.5	85.4	100.0	100.0	0.0 (5)	100.0	0.0
Compressed	100.0	0.0	0.0	100.0	98.6	1.4	0.0 (5)	100.0	0.0
Non-invariant (income)	100.0	2.9	0.0	100.0	100.0	0.0	100.0 (25)	0.0	0.0

Note. Percentage of readings that pass, except the columns marked fail or unresolved. Gate, L2 and L3 are read per pseudo-model (280 readings per type); L1, R1 and R2 on one pseudo-model per replicate (70 replicates). R4 is read at margin .08 on the number of replicates in parentheses; it gives no verdict without a general factor. L2 passes in no reading of any type; its remaining readings are unresolved. L4 is the level-4 pass (R1, R2 and R4 in both framings). Shifted types raise every persona by the stated points; the doubled gap raises low-income personas only; non-invariant makes three items independent within the low-income group. Construction, seeds and per-step results: `paper_brm/analysis_brm/intermediate_panel` in the release.

Real people are rarely failed, and planted failures are caught. Real respondents pass the gate and level 3 in every reading, level 1 in 92.9% of replicates, and R1 and R2 of level 4 in every replicate, and level 2 fails them in no reading (Table 3). At 30 draws real respondents' persona means reach a dependability of only .70, because real demographic cells differ little from one another. Applied to every study, a floor of $\phi \geq .80$ would fail real people, and the gate therefore requires it only of studies that compare or rank individual personas. Each planted failure trips the rung it targets. Independent items fail level 1, R1 and R2 in every replicate, and compressed samples fail level 1 in every replicate. Shifted samples fail level 3 in every reading, and non-invariant samples fail R4 in

all 25 replicates. A doubled income gap fails level 2 in 92.5% of per-model readings. Two rungs also react to failures they do not target. A sample shifted by 2.1 points passes the gate in only 28.6% of readings, as more depressed respondents vary more between draws. R2 also fails compressed and non-invariant samples, both of which change the size of the loadings. Over a further 200 replicates with graded doses, each rung detects in 95% of replicates a distortion smaller than the one the language models show at levels 1, 3 and 4 (Supplement S5). In those replicates real respondents' misfit ratio averages 1.15, and level 1 passes them at $\tau = 1.5$ in 99.4% of 800 per-model readings, which sets the level-1 floor.

What level 2 and R4 need to certify a pass. A group gap is KEPT when the whole interval for the ratio of the simulated gap to the real gap lies between 0.75 and 1.25, placing the simulated gap within a quarter of the real gap's size. Because the whole interval must fall inside that range, a kept verdict depends on the precision of both gap estimates as well as on how close they are. Certifying a kept gap therefore needs more data than detecting a gap that is not kept, and level 2 and the invariance rule R4 both carry this asymmetry. A second simulation repeated the control panel over 60 replicates on grids of 48, 144 and 432 persona cells. Level 2 fails a faithful simulator built from real respondents in at most 3.3% of per-model readings and fails a doubled income gap in 97.5% to 100%. At 30 draws per persona the low-to-high income gap of a faithful simulator is never read KEPT, because draws of one persona vary enough to hold its interval wider than the kept region. With no simulation error, the limit of adding draws, the same gap is read KEPT in 95.0% and 98.3% of replicates on the 48-cell and 144-cell grids. The Black–White and Hispanic–White gaps are never read KEPT on any grid, because NHANES measures them too loosely. A researcher who needs a kept gap therefore plans for more draws per persona and for a population gap the survey estimates with a narrow interval, and a finer persona grid does not supply the second. R4 shows the same asymmetry, and at the .08 margin, 24 to 40 personas per group leave some invariance steps unresolved for real respondents in 24 of 25 replicates, while a planted invariance failure fails R4 in every replicate. A design that

needs an R4 pass needs more personas per group than the corpus has.

Three External Datasets

The ladder was applied without changing any rule to three silicon-sample datasets released by other authors. For each dataset we report which rungs the release allows and what the verdicts tell a researcher who would use that sample. Table 4 gives the rungs each release supports, and Figure 3 sets the level-2 verdicts beside those of the worked example.

Table 4

Rungs Each Dataset Supports, and the Verdicts

Dataset	Gate	Level 1	Level 2	Level 3	Level 4
Worked example	Pass	Fail	Fail, 0 of 28 kept	Fail	Fail
Bisbee et al.	Pass, 11 of 11	Not run (a)	Fail, 3 of 77 kept	7 of 11 pass	Not run (a)
Argyle et al.	Not run (b)	Not run (b)	Fail, 2 of 61 kept	Illustration (d)	Not run (a)
OpinionQA	Not run (c)	Not run (c)	Fail, 1 of 154 kept	Illustration (d)	Not run (c)

Note. Bisbee et al. under the full prompt, with gate and level-3 counts over the 11 thermometers; Argyle et al. Study 3, main run. A rung is not run when the release lacks what it needs: (a) single-item outcomes, with no multi-item scale; (b) one answer per respondent, with no repeated draws or answer vectors; (c) group answer distributions only, with no individual answers. (d) Level 3 needs a use tolerance the original authors did not state, so it appears only as an illustration in Supplement S6.

Bisbee et al.: ChatGPT Feeling Thermometers

Bisbee et al. (2024) gave ChatGPT (gpt-3.5-turbo, June 2023) the profiles of 7,530 respondents to the 2016 and 2020 American National Election Studies and asked, about 30 times under each of three prompts, for a feeling thermometer toward each of 11 groups. Because every persona has a human twin, the same respondents’ own thermometers serve as the reference, and the release supports the gate, level 2 and level 3. Our cleaning of the released file reproduces the authors’ analysis data and their published regression to six

decimals. The gate passes for every thermometer, with at most 11 draws needed under the recommended rule, and level 2 fails under every prompt. Under the full prompt, 3 of the 77 contrasts are kept, among them the partisan gaps in feeling toward the Democratic and Republican parties (ratios 1.12 and 1.05), both large and precisely measured in the survey. The Republican–Democrat gap is steepened toward five other groups, and several gaps by sex and age are reversed, missing or attenuated. Level 3 passes at half a standard deviation for 7 of the 11 thermometers under the full prompt. For a researcher, these verdicts support using the sample for the partisan gaps toward the two parties and for the average level of most thermometers, and they rule out using it for partisan feeling toward other groups or for differences by sex and age. Bisbee et al. (2024) already read these data with caution, and the ladder places that caution on the specific uses that fail.

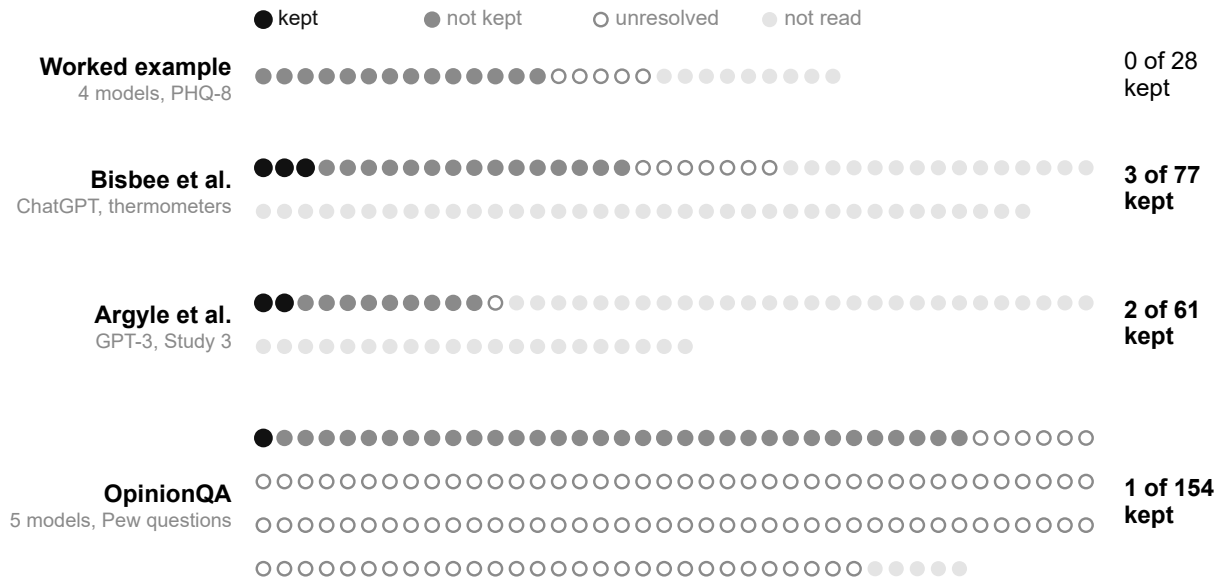
Argyle et al.: GPT-3 Answering for Survey Respondents

In their Study 3, Argyle et al. (2023) gave GPT-3 each of 4,270 American National Election Studies respondents’ answers to 11 of 12 interview questions and asked it to answer the twelfth. The study tested pattern correspondence, the criterion of their algorithmic fidelity under which simulated answers reproduce the relationships among attitudes, demographics and behaviour that people show. Holding one answer per respondent and outcome, the release cannot support the gate or level 1, and level 2 compares groups of the same respondents. Of 61 contrasts in the main run, 2 are kept, the Republican–Democrat gaps in ideology and in Trump’s two-party vote share (both 0.92). Nine are not kept, among them the Black–White gap in Trump’s vote share, which GPT-3 reproduces at about half its size (0.53). Most of the remaining contrasts are not read, because the survey’s subgroups are too small to measure their gaps. The verdicts support the partisan gaps, where the original claim is strongest, and leave the racial and other demographic gaps unsupported, narrowing the claim of algorithmic fidelity to the partisan comparisons. Lacking a use tolerance from the original authors, level 3 for this release appears only as an illustration in Supplement S6.

OpinionQA: What a Crude Reference Allows

Meister et al. (2025) steered five models toward demographic groups on 100 contested questions from the Pew American Trends Panel and released the models' answer distributions beside the Pew distributions for the same groups (Santurkar et al., 2023). The human side is each group's distribution of answers to a question, built from unweighted counts, with no individual respondents. Without repeated draws of a persona or individual answer vectors, the gate, level 1 and level 4 cannot run. With the model steered on one attribute, such as being a Democrat, while the Pew groups also differ in age, income and every other attribute, the reference gives a crude gap in place of the conditional gap the steering asks about. Averaging each contrast over questions with differing gaps also leaves its interval wide.

Of the 154 contrasts across models, elicitation methods and steering methods, 1 is kept, 33 are not kept, 115 are unresolved and 5 are not read. A comparison of point estimates would credit far more fidelity than the data establish, with the point ratio inside the kept region for 37 of the 149 contrasts read. The ladder keeps 1 of those 37 and reports the other 36 as unresolved, and a researcher using these releases can claim little about group gaps in either direction.

Figure 3*Level-2 Group Gaps in the Worked Example and Three External Datasets*

Note. Each dot is one group gap, read per model in the worked example and OpinionQA and per thermometer or outcome in the releases of Bisbee et al. (full prompt) and Argyle et al. (main run). A gap is kept when its interval lies between 0.75 and 1.25 times the real gap, not kept when it lies wholly outside that range, unresolved when it overlaps it, and not read when the reference cannot tell the gap from zero or measures it too imprecisely. Intervals are Bonferroni-corrected within each model, prompt or run.

What the Three Datasets Show

Applied to other authors' samples, the ladder separates the uses a release supports from the uses it does not, and it reports where the human data are too thin to decide. Repeated draws of each persona under a fixed prompt make the gate and level 1 possible, and individual answer vectors make level 4 possible. Personas built from real respondents' profiles, as in the releases of Bisbee et al. (2024) and Argyle et al. (2023), give a reference that matches the simulated contrast. A large, individually measured reference puts a kept verdict within reach for the gaps a study targets. Using a crude, group-level reference, level

2 detects broken gaps, and a kept verdict stays out of reach.

Discussion

In both the worked example and the three external datasets, each sample failed at least one check tied to a use researchers make of such samples. In the worked example none of these failures could be seen in a single answer. Our recommendation for researchers running persona studies is to begin before the generation step, first by establishing whether human data exist to ground the sample, and second by deciding which claims about the population the study intends to make. Holding the model’s ability to stand in for those people in critical doubt, in good faith, the researcher then checks each claim against the rung designed to test it. Group gaps, population levels, single cases and scale scores are separate extrapolations, and each one needs its own check.

Using the ladder depends on meeting its requirements. Every level above the gate compares the simulated sample against human data, and without a reference on the same instrument the ladder cannot be applied. Each rung additionally carries its own requirements, set out in Figure 1; where a study’s data cannot meet them, that rung is not run and the claims it would support are not made. Because the reference determines which personas can be compared, how many draws each needs and which gaps can be measured precisely enough to certify, it has to be in hand before generation. Additionally, the researcher fixes how each persona attribute maps to a reference variable before generation and reports that mapping with the verdicts, because in the worked example matching income on the label’s dollar amount rather than the poverty-ratio band changed the verdict on one contrast. Personas built on real respondents’ profiles, as Bisbee et al. (2024) and Argyle et al. (2023) did, offer the cleanest reference by giving every simulated answer a human twin.

Our controls and the three external datasets address whether the ladder itself can be trusted. Pseudo-models built from real respondents rarely fail, while failures planted into those same respondents are caught at the rung each one targets. Applied to other authors’

releases without changing a rule, the ladder ran on instruments, models and designs it was not built around. In those releases it supports the partisan gaps reported by Bisbee et al. (2024) and Argyle et al. (2023), while most gaps by sex, age and race either fail or cannot be read. We do not dispute those studies' findings, though the results point toward a narrower interpretation of their claims of fidelity, centred on the partisan comparisons.

In practice the ladder strengthens the context around each claim drawn from a persona sample and weeds out readings the data do not support. The right value for a gap or a level still has to come from the human data.

Regarding the ladder's own limitations, its thresholds were fixed before the external analyses and act as defaults, to be replaced by a tolerance a user can justify for their own use. Certifying a pass also takes more draws, more personas per group and a more precise reference than detecting a failure, and an unresolved verdict can reflect the design as well as the simulator. Under the design of the worked example, with 30 draws per persona and a national survey as reference, level 2 serves mainly to detect distorted gaps. Finally, the frozen ladder does not read contrasts with no population gap, leaving untested any gap a model invents where people show none; an equivalence test on the simulated gap in raw units would cover this case.

Conclusion

The validity ladder replaces a single judgement of whether a simulated sample resembles people with a separate check for each use of the sample, each carrying a pass rule, a human reference and the claim it supports. Across 28,800 PHQ-8 assessments from four models, the ladder read every model's persona means and failed every model on each use above them. Applied to three published silicon samples, it narrowed what each release can be used to claim and identified claims some releases cannot test at all. For a researcher, the result is a list of the claims a sample supports, each tied to the check behind it. We release the code for every rung and the receipt for every number, allowing grounded or fine-tuned simulators to be checked against the same baseline.

Declarations

Funding. No funding was received for this work.

Competing interests. The author has no competing interests to declare.

Ethics approval. Not applicable. The study analyses public, de-identified NHANES microdata and text generated by language models. No human participants were recruited.

Consent to participate and for publication. Not applicable.

Availability of data, materials and code. The corpus, the persona registries, the original generation code for both framings, the analysis scripts and every receipt file are at <https://github.com/pskeough/A-Validity-Ladder-for-LLM-Simulated-Populations>. NHANES data are public from the National Center for Health Statistics, and the release includes the code that extracts them.

Author contributions. Patrick Keough is the sole author and takes responsibility for all content.

Generative AI disclosure. Generative AI tools (Anthropic’s Claude, through Claude Code) were used for language editing, for LaTeX formatting, and to write the Python analysis and verification scripts to the author’s specification. The research questions, study design, statistical specification and interpretations are the author’s own, and all generated code was reviewed by the author. The verification scripts that check every numeric claim in this manuscript against its receipt are released with the data.

Open Practices Statement

The data, materials and analysis code for this study are available at <https://github.com/pskeough/A-Validity-Ladder-for-LLM-Simulated-Populations>. The analysis reported here is release commit 2550a58. Supplement S2 maps every number in the body of the article to the file in that repository that prints it and the script that writes the file. The ladder’s rules were set on the worked example, run once on BFI-44

answers from three models (Jiang et al., 2024) against the Twin-2K-500 panel (Toubia et al., 2025), and frozen in `LADDER_SPEC.md` at commit `91a1b10` (2 October 2026), before any external dataset was analysed; the corpus analyses were not registered in advance. The decoding control’s design file, dated before its first call, is in the repository.

References

- Argyle, L. P., Busby, E. C., Fulda, N., Gubler, J. R., Rytting, C., & Wingate, D. (2023). Out of one, many: Using language models to simulate human samples. *Political Analysis*, 31(3), 337–351. <https://doi.org/10.1017/pan.2023.2>
- Bisbee, J., Clinton, J. D., Dorff, C., Kenkel, B., & Larson, J. M. (2024). Synthetic replacements for human survey data? The perils of large language models. *Political Analysis*, 32(4), 401–416. <https://doi.org/10.1017/pan.2024.5>
- Brennan, R. L. (2001). *Generalizability theory*. Springer.
<https://doi.org/10.1007/978-1-4757-3456-0>
- Chen, T.-C., Parker, J. D., Clark, J., Shin, H.-C., Rammon, J. R., & Burt, V. L. (2018). *National Health and Nutrition Examination Survey: Estimation procedures, 2011–2014* (Vital and Health Statistics, Series 2 No. 177). National Center for Health Statistics. https://www.cdc.gov/nchs/data/series/sr_02/sr02_177.pdf
- Cronbach, L. J., Gleser, G. C., Nanda, H., & Rajaratnam, N. (1972). *The dependability of behavioral measurements: Theory of generalizability for scores and profiles*. New York: Wiley.
- Desai, M., Card, D., & Jacobs, A. Z. (2026). *Validating LLMs in social science: Epistemic threats and emerging norms*. arXiv preprint arXiv:2607.07915.
<https://doi.org/10.48550/arXiv.2607.07915>
- Drasgow, F., Levine, M. V., & Williams, E. A. (1985). Appropriateness measurement with polychotomous item response models and standardized indices. *British Journal of Mathematical and Statistical Psychology*, 38(1), 67–86.
<https://doi.org/10.1111/j.2044-8317.1985.tb00817.x>
- Fieller, E. C. (1954). Some problems in interval estimation. *Journal of the Royal Statistical Society: Series B (Methodological)*, 16(2), 175–185.
<https://doi.org/10.1111/j.2517-6161.1954.tb00159.x>
- Holt, D., & Smith, T. M. F. (1979). Post stratification. *Journal of the Royal Statistical*

- Society. Series A (General)*, 142(1), 33–46. <https://doi.org/10.2307/2344652>
- Jiang, H., Zhang, X., Cao, X., Breazeal, C., Roy, D., & Kabbara, J. (2024). PersonaLLM: Investigating the ability of large language models to express personality traits. In *Findings of the association for computational linguistics: NAACL 2024* (pp. 3605–3627). Mexico City, Mexico: Association for Computational Linguistics. <https://aclanthology.org/2024.findings-naacl.229>
- Johnson, C. L., Paulose-Ram, R., Ogden, C. L., Carroll, M. D., Kruszon-Moran, D., Dohrmann, S. M., & Curtin, L. R. (2013). *National Health and Nutrition Examination Survey: Analytic guidelines, 1999–2010* (Vital and Health Statistics, Series 2 No. 161). National Center for Health Statistics. https://www.cdc.gov/nchs/data/series/sr_02/sr02_161.pdf
- Keough, P. (2026). *Plausible patients, impossible populations: Auditing epidemiological fidelity in large language model mental health simulations*. arXiv preprint arXiv:2604.17359. (Version 2, 5 August 2026) <https://doi.org/10.48550/arXiv.2604.17359>
- Kroenke, K., Strine, T. W., Spitzer, R. L., Williams, J. B. W., Berry, J. T., & Mokdad, A. H. (2009). The PHQ-8 as a measure of current depression in the general population. *Journal of Affective Disorders*, 114(1–3), 163–173. <https://doi.org/10.1016/j.jad.2008.06.026>
- Lakens, D., Scheel, A. M., & Isager, P. M. (2018). Equivalence testing for psychological research: A tutorial. *Advances in Methods and Practices in Psychological Science*, 1(2), 259–269. <https://doi.org/10.1177/2515245918770963>
- Lin, Z. (2026a). Large language models as psychological simulators: A methodological guide. *Advances in Methods and Practices in Psychological Science*, 9(1), Article 25152459251410153. <https://doi.org/10.1177/25152459251410153>
- Lin, Z. (2026b). A validity-guided workflow for robust large language model research in psychology. *Behavior Research Methods*, 58(8), Article 216.

<https://doi.org/10.3758/s13428-026-03073-2>

Lorenzo-Seva, U., & ten Berge, J. M. F. (2006). Tucker's congruence coefficient as a meaningful index of factor similarity. *Methodology*, 2(2), 57–64.

<https://doi.org/10.1027/1614-2241.2.2.57>

Meister, N., Guestrin, C., & Hashimoto, T. (2025). Benchmarking distributional alignment of large language models. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)* (pp. 24–49). Association for Computational Linguistics. <https://doi.org/10.18653/v1/2025.naacl-long.2>

National Center for Health Statistics. (2018). *National Health and Nutrition Examination Survey: Analytic guidelines, 2011–2014 and 2015–2016*. Centers for Disease Control and Prevention. <https://wwwn.cdc.gov/nchs/data/nhanes/analyticguidelines/11-16-analytic-guidelines.pdf> (Dated December 14, 2018)

Olsson, U. (1979). Maximum likelihood estimation of the polychoric correlation coefficient. *Psychometrika*, 44(4), 443–460. <https://doi.org/10.1007/BF02296207>

Park, P. S., Schoenegger, P., & Zhu, C. (2024). Diminished diversity-of-thought in a standard large language model. *Behavior Research Methods*, 56(6), 5754–5770. <https://doi.org/10.3758/s13428-023-02307-x>

Samejima, F. (1969). Estimation of latent ability using a response pattern of graded scores. *Psychometrika*, 34(S1), 1–97. <https://doi.org/10.1007/BF03372160>

Santurkar, S., Durmus, E., Ladhak, F., Lee, C., Liang, P., & Hashimoto, T. (2023). Whose opinions do language models reflect? In *Proceedings of the 40th International Conference on Machine Learning* (Vol. 202, pp. 29971–30004). PMLR. <https://proceedings.mlr.press/v202/santurkar23a.html>

Savalei, V., Brace, J. C., & Fouladi, R. T. (2024). We need to change how we compute RMSEA for nested model comparisons in structural equation modeling. *Psychological Methods*, 29(3), 480–493. <https://doi.org/10.1037/met0000537>

- Schirmann, D. J. (1987). A comparison of the two one-sided tests procedure and the power approach for assessing the equivalence of average bioavailability. *Journal of Pharmacokinetics and Biopharmaceutics*, 15(6), 657–680.
<https://doi.org/10.1007/BF01068419>
- Sinharay, S. (2016). Asymptotically correct standardization of person-fit statistics beyond dichotomous items. *Psychometrika*, 81(4), 992–1013.
<https://doi.org/10.1007/s11336-015-9465-x>
- Snijders, T. A. B. (2001). Asymptotic null distribution of person fit statistics with estimated person parameter. *Psychometrika*, 66(3), 331–342.
<https://doi.org/10.1007/BF02294437>
- Toubia, O., Gui, G. Z., Peng, T., Merlau, D. J., Li, A., & Chen, H. (2025). *Twin-2K-500: A dataset for building digital twins of 2,000 people*. arXiv preprint arXiv:2505.17479.